

Tiffany Lee

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Summary

Researcher in computational and cognitive neuroscience interested in how neural dynamics and representational structures give rise to cognition through **neural manifold**, **state-space analysis**, and **theoretical modeling**.

Education

Gwangju Institute of Science and Technology (GIST) Mar 2022 – Dec 2026

B.S. in Electrical Engineering and Computer Science

Minor in Biomedical Science and Engineering

- **GPA:** 3.88/4.0, *Magna Cum Laude (expected)*
- **Key Courses:** Linear Algebra, Discrete Mathematics, Bio Statistics and Machine Learning, Introduction to Artificial Intelligence, Computational Models of Cognition, Information Theory

University of California, Berkeley

Jan 2025 – May 2025

Berkeley Global Access Program

Korea Advanced Institute of Science and Technology (KAIST)

Aug 2024 – Dec 2024

Visiting Student

University of California, Berkeley

June 2023 – Aug 2023

Summer Session Program

Research Experience

Neurophotronics Lab, GIST

Gwangju, South Korea

Undergraduate Researcher (Advisor: Prof. Euiheon Chung)

Sep 2025 – Present

- Led a project on spectral graph theoretical analyses on primary cortical neuron cultures using high-density multielectrode arrays (HD-MEAs) with amyloid-beta oligomers.
- Led a project investigating the impact of trial-to-trial shared neural variability and its alignment with the readout axis on the cognitive decline in older adults.

The Computation and Language Lab, UC Berkeley

Berkeley, United States

Research Assistant (Advisor: Prof. Steven Piantadosi)

May 2025 – Present

- Led a project decoding causal states from feature vectors in task-trained recurrent neural networks (RNNs) for cognitive modeling.
- Investigated how causal states can function as general cognitive states, and analyzed their embedding properties in functional magnetic resonance imaging (fMRI) and RNNs.

Brain Machine Intelligence Lab, KAIST

Daejeon, South Korea

Undergraduate Researcher (Advisor: Prof. Sang Wan Lee)

Sep 2024 – Dec 2024

- Participated in a project decoding electroencephalogram (EEG) signals of prediction errors in model-based and model-free decision-making.
- Assisted graduate students with literature review, code evaluation, and research discussions.

BioComputing Lab, GIST

Gwangju, South Korea

Undergraduate Researcher (Advisor: Prof. Sung Chan Jun)

June 2024 – Aug 2024

- Led a project exploring behavioral and EEG responses in Deepfake face discrimination under label alteration.
- Designed and conducted an EEG recording behavioral experiment and analyzed event-related potentials.

Neurophotronics Lab, GIST

Gwangju, South Korea

Undergraduate Researcher (Advisor: Prof. Euiheon Chung)

Dec 2022 – Feb 2024

- Participated in a HD-MEA project, focusing on research design and the initial experiment setup.

Publications

- Lee, H., & Piantadosi, S. T. (in preparation). Modeling Internal Representations of Task-Trained RNNs as Cognitive-State Automata.
- Lee, H., Ashiquzzaman, A., Kwon, H. S., Kim, Y. R., Kim, S. S., & Chung, E. (in preparation). Spectral Graph Theoretical Characterization of Cortical Neuronal Networks using HD-MEA.

Conferences

- Lee, H., & Chung, E. (2026, May 7-9). *Influence of Hierarchical Task Inference on Representational Dynamics in Healthy and Aging-like RNNs under Hidden-Rule Conditions* [Poster]. The Korean Society of Medical & Biological Engineering Spring Annual Meeting, Jeju, South Korea.
- Lee, H., & Chung, E. (2025, Nov 8). *Age-Related Reorganization of Shared Neural Variability and Readout Alignment During Cognitive Control* [Poster]. The 19th International Symposium for Aging, Gwangju, South Korea.
- Lee, H., Ashiquzzaman, A., Kwon, H. S., Kim, Y. R., Kim, S. S., & Chung, E. (2025, Nov 6-8). *Spectral Graph Theoretical Characterization of Cortical Neuronal Networks using HD-MEA* [Poster]. The Korean Society of Medical & Biological Engineering Fall Annual Meeting, Gimhae, South Korea.

Teaching Experience

Single Variable Calculus and Applications: Teaching Assistant	<i>Mar 2026 – June 2026</i>
Computer Programming: Teaching Assistant	<i>Mar 2026 – June 2026</i>
PIUM: Science Tutoring Volunteer for the Suburbs	<i>Sep 2022 - Nov 2022</i>

Extracurricular Activities

AlleoDreamy: AI Newsletter Team	<i>Nov 2024 – Aug 2025</i>
Google Developer Student Clubs: External Relations Leader	<i>Sep 2023 – Feb 2024</i>
Buddy Program: Volunteer for International Freshmen	<i>Sep 2023 – Dec 2023</i>
Korean I-Corps: Mock Startup Team	<i>Jun 2022 – Feb 2023</i>
Asian Science Camp: Representative of Korea	<i>Jun 2022 – Jun 2022</i>

Skills

Experiment: Primary Neuron Culture, HD-MEA, EEG, Vibratome, Mouse Behavior Test, Calcium Imaging

Software: Brainwave (3Brain), EEGLAB, UCINET

Programming: Python, MATLAB, C, R, Java

Language: Korean (Native), English (Fluent; TOEFL 113/120)

Awards & Honors

GIST-Fellowship for Innovative Research in Science & Technology	<i>June 2026</i>
Best Poster Award <i>The Korean Society of Medical & Biological Engineering Spring Annual Meeting</i>	<i>May 2026</i>
National Scholarship for Science and Technology	<i>Mar 2022 – Dec 2025</i>
Best Poster Award <i>The 19th International Symposium for Aging</i>	<i>Nov 2025</i>
Study Abroad Program Scholarship	<i>Jan 2025 – Mar 2025</i>
Academic Excellence Scholarship	<i>Sep 2024 – Dec 2024</i>
AWS AI Award (1st place/200+ teams nationwide) <i>AI4GOOD Hackathon</i>	<i>Mar 2024</i>
Academic Excellence Scholarship	<i>Mar 2022 – Dec 2023</i>